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SANTA CLARA UNIVERSITY	ELEN 164 Fall 2023	Shoba Krishnan
Laboratory #8: Stability of the Buck Converter		

I. OBJECTIVES

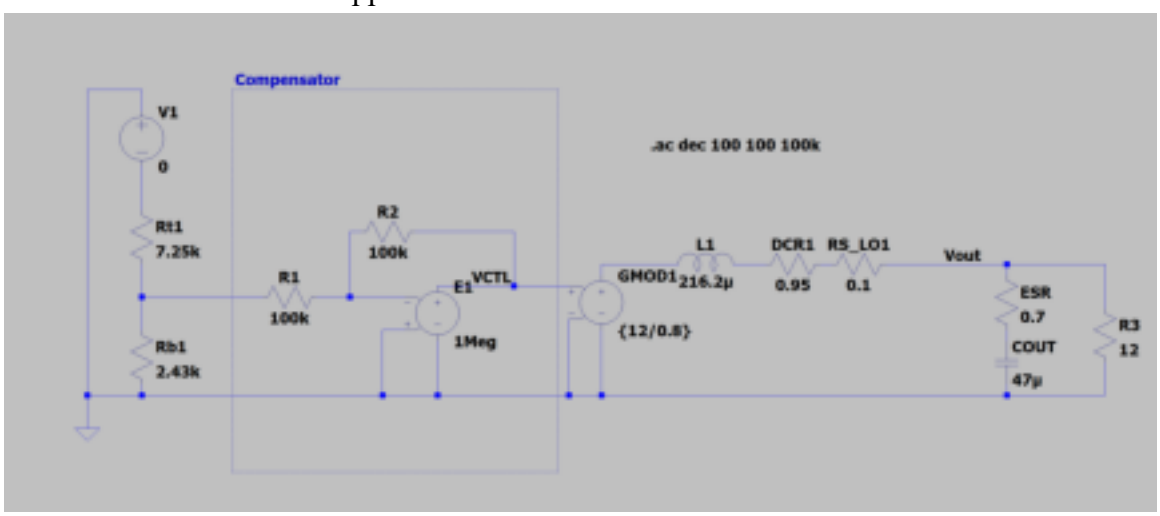
- To study the closed loop behavior of the Buck converter.
- To build a compensator to stabilize the converter.
- To observe the stability through the transient behavior of the converter. **III.**

Laboratory Procedure

Part I: AC response

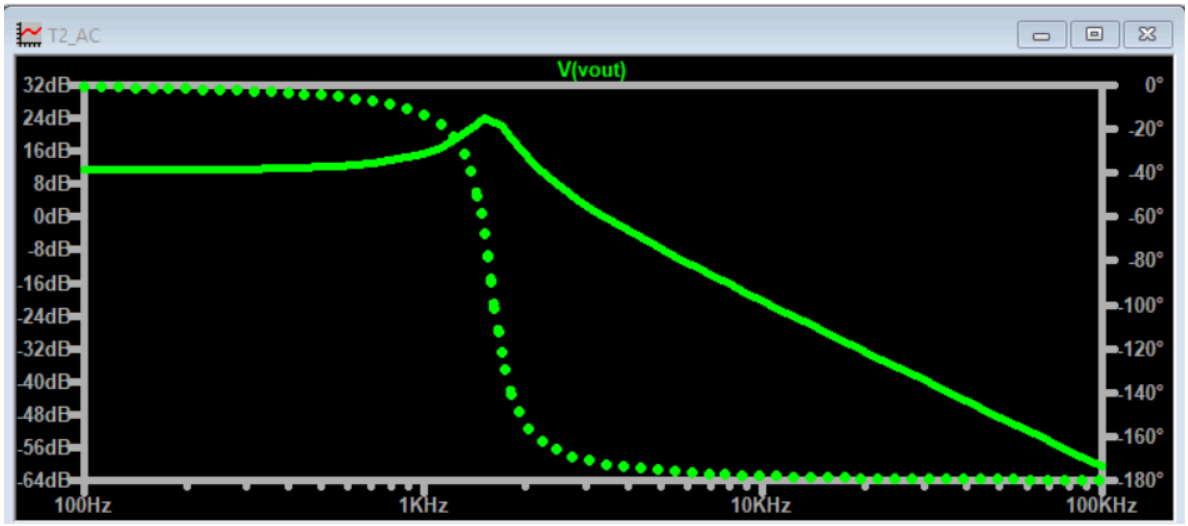
1) Download the T2_AC.asc file from the Camino, under Lab 8 module. 2) Open up the file in LTSpice and save this circuit as a new file.

The below schematic will appear.



The Rt1 and Rt2 voltage divider represents the feedback network. The gain on the two voltage-controlled voltage sources correspond to the gain of the Converter and the ramp function of the PWM circuit.

3) Set the DCR of L1 (.95 def) to zero and ESR (.7 def) of COUT to zero and run the simulation to get a bode plot showing the transfer function of the converter. Ensure that you are observing until at least 10 to 50 MHz on the x axis.



Observations:

a) What is the general shape of this plot and what type of system is it representing?

This bode plot shows an ideal buck converter without an fESR zero.

Does this transfer function indicate the presence of poles and zeroes?

Yes, two poles at f0 (~1.6kHz). No zeroes on this ideal system tho.

b) What is the DC gain?

It's Vout/Vin which is 0dB

Why is there high gain at DC?

The gain is "high" because the converter acts like a low pass filter. Don't get it twisted though, the gain is only high relative to the rest of the graph.

c) Where does the gain start to drop?

It drops at the cutoff frequency.

Give the formula that gives the frequency at which this occurs.

F0 = 1/(2 * pi sqrt(LC))

What is the slope at this point of the magnitude bode plot?

-40dB because there are two poles.

What is the purpose for this drop in gain and steep slope?

To reject high frequencies.

At this point where the gain drops, what is the phase angle of the loop?

-90

d) What is the gain at the switching frequency?

About -40dB.

e) Where does the gain cross the 0dB axis?

About 3.4Khz.

What is the phase angle at that point?

About -170 degrees.

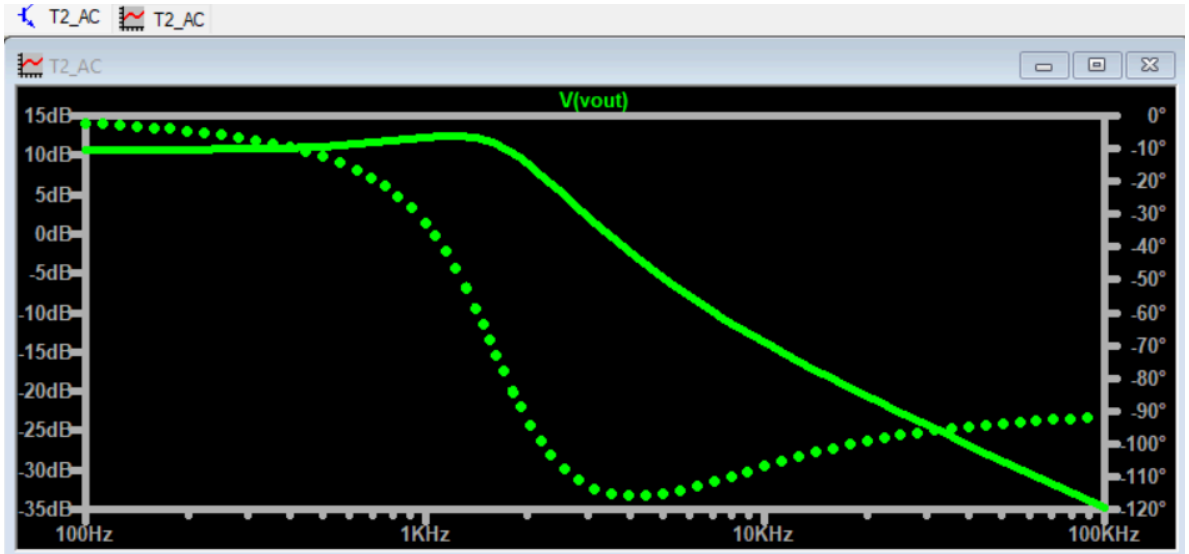
What is the phase margin for the system?

Roughly 10 degrees.

4) Now make the DCR 0.95 ohms and run the simulations.

What do you notice on the plot? Explain why.
Not as large of an f0 spike. The resistor helps lessen this.

5) Now change the ESR to 0.7 ohms and run the simulations again.



a) Where does the gain start to drop?
Same spot.
Give the formula that gives the frequency at which this occurs.
Same as before
What is the slope at this point of the magnitude bode plot?
What is the purpose for this drop in gain and steep slope?
-40dB/dec
At this point where the gain drops, what is the phase angle of the loop?
-90
b) At what frequency does the next change of gain happen?
At fESR (4.8kHz). It changes to -20dB
Why does this occur?
There's a zero here.
Give the formula that gives this point.
fESR = 1/ (2 * pi * C * ESR)
c) What is the gain at the switching frequency?
About -24.55dB
Explain if this change helps when taking the switching frequency into consideration?
It makes it worse because the gain is higher here now.

d) Where does the gain cross the 0dB axis?
About 3.38kHz.
What is the phase angle at that point?

About -114 degrees.
What is the phase margin for the system?
About 66 degrees.

6) We will be comparing the original AC response to the compensated response, so we need to save the original file separately.

7) Design a compensator for this converter (with ESR included) assuming a cross-over frequency of 5KHz (⅙ of fs) and a phase margin of 45 degrees.

Analysis:
Show all steps of the compensator design. Use the K factor method to identify the type of compensator and then calculate the component sizes needed to achieve the compensator.
Include your calculations in the lab report.

Lab 8 Design portion

Type 2

phase margin of 45 means min phase = $-180+45$
$= -135$

zero at $\frac{1}{2\pi R_2 C_1}$ $\quad G_c = -114$ $\quad \theta_{comp}$ $114+45 = 159°$ $\approx 160°$

pole at $\frac{1}{2\pi R_2 C_2}$

$f_{co} = 5\,kHz$, at this, we get $-5.553\,dB$

$-5.553 + 20\log G_{comp}(\ ) = 0$

$G_{comp} = 1.8225\,V$

$G = \frac{R_2}{R_1} = \frac{180k}{100k} \rightarrow$ choose

$K = \tan\left(\frac{160}{2}\right) = 5.671$

$\omega_{co} = f_o 2\pi = 5k(2\pi) = 31415.927$ rad/s

$\omega_z = \frac{\omega_{co}}{K} = 5539.475 \rightarrow 881.635\,Hz$

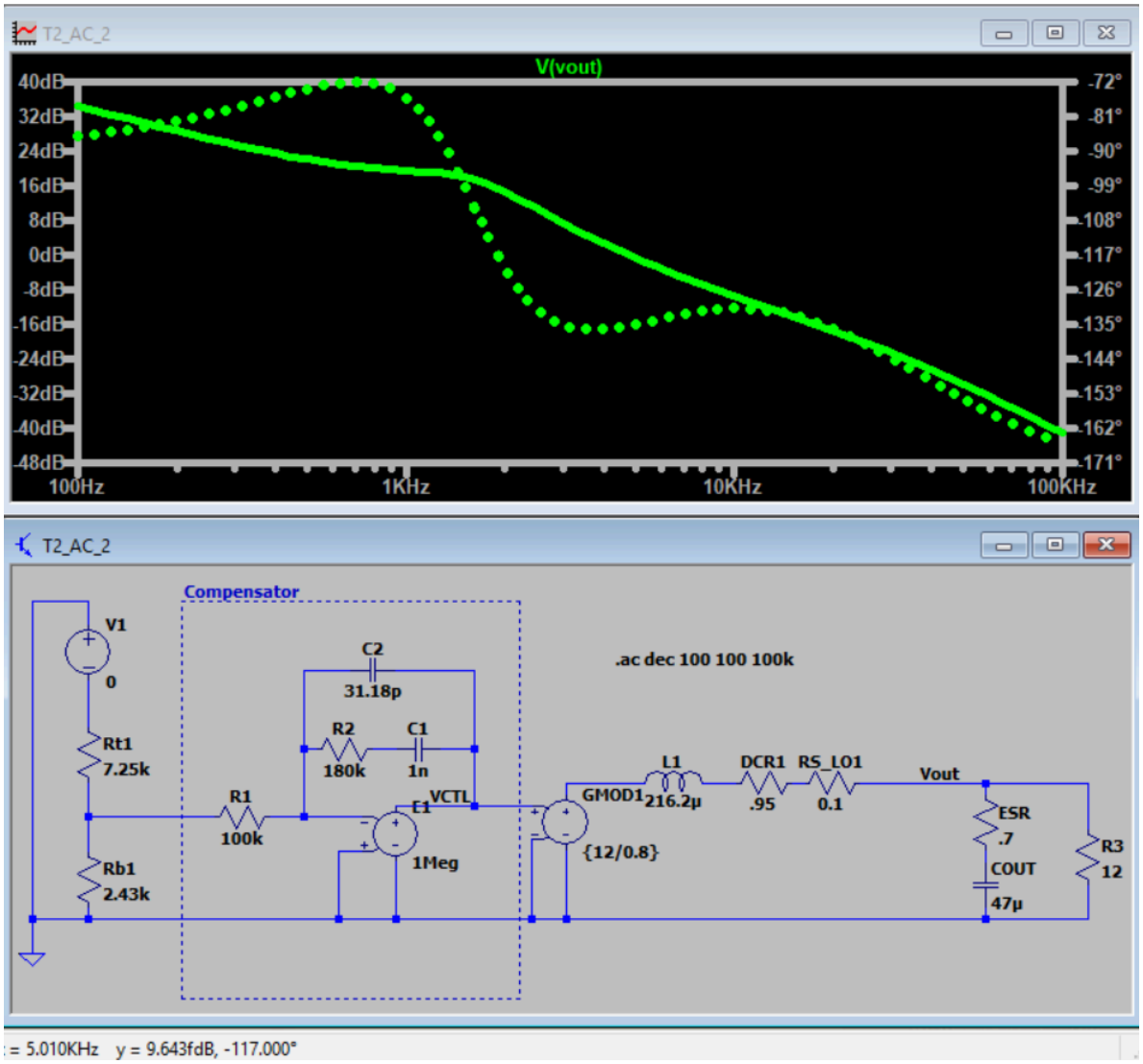
$\omega_p = K\omega_{co} = 178168.573 \rightarrow 28356.41\,Hz$

$C_1 = \frac{K}{2\pi f_{co} R_2} = 1\,nF$

$C_2 = \frac{1}{K 2\pi f_{co} R_2} = 31.18\,pF$

8) Add in your compensator design in the box labeled “compensator” and run the

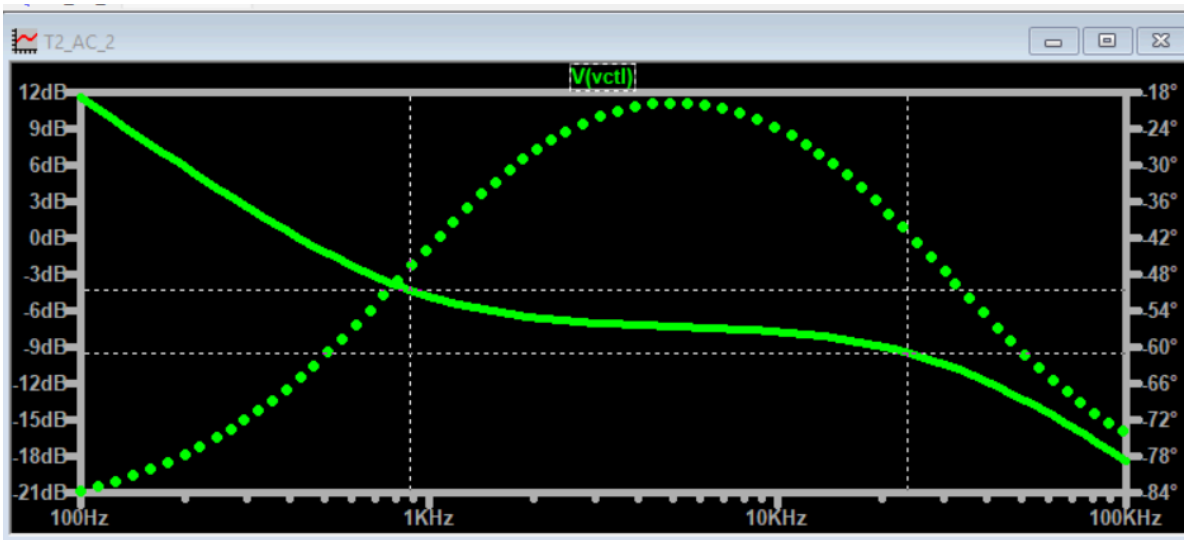
simulation. You will see a solid line showing the gain, and a dotted green line showing the phase of each node you select.

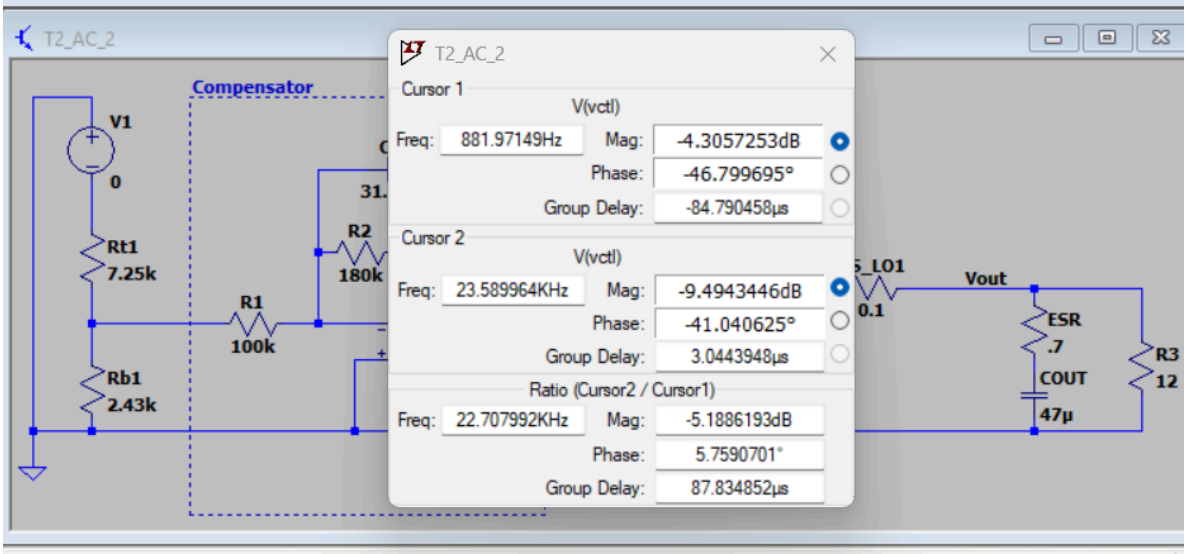


9) Compare the original design's gain and phase graphs with the new compensated version. You can probe Vout to see the bode plot of the converter, and VCTL to probe the bode plot of just the compensator.

We can see the crossover is now roughly at 5kHz and the phase is -135 degrees.

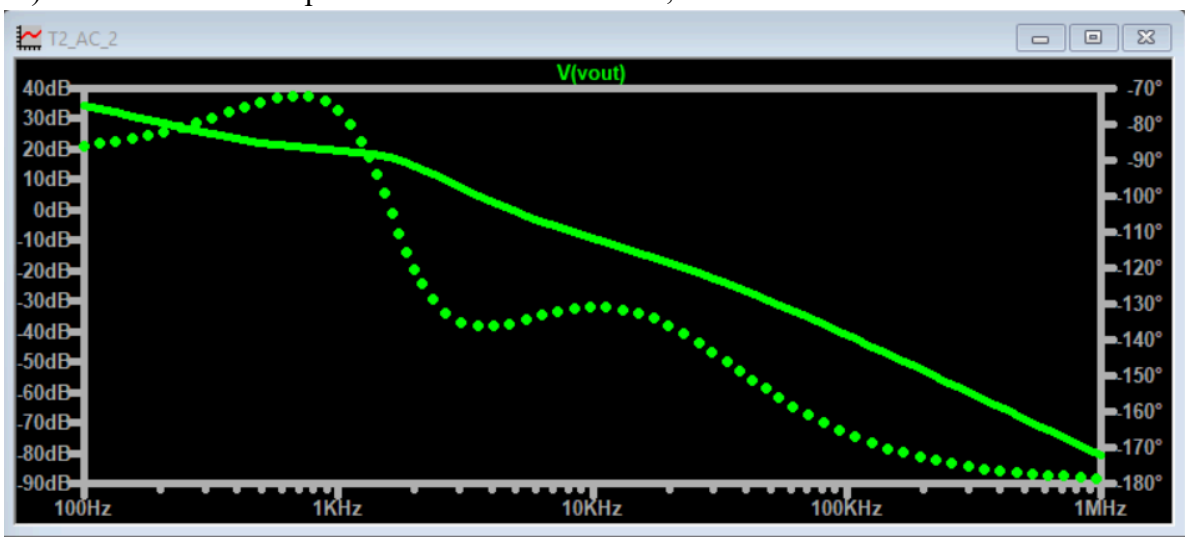
10) From the Bode Plot of the compensator, use cursors to locate the relative positions of zeros and poles you designed in. Do they match with your calculated values?





These are close to the calculated values, but the pole is a little off.

11) From the closed loop Bode Plot of the converter, find the



a. phase margin

The phase margin at crossover is 45 degrees (good)

b. gain margin

The gain is 0 at the crossover freq.

12) Comment on the stability of the system

The system is relatively stable. If gain was positive, it would be more stable.

13) Note the value of the magnitude of the closed loop transfer function at the switching frequency.

It is roughly -23dB.